Blockchain Developer Take-Home Assignment

Overview

Build a **Digital Certificate Issuance System** using Hyperledger Fabric that demonstrates your blockchain development skills, smart contract implementation, and testing capabilities.

Time Estimate: 3-4 hours
Submission Deadline: 3 days from receipt

Business Scenario

Create a permissioned blockchain system for issuing and verifying educational certificates where:

- **Universities** issue certificates
- **Students** receive and own certificates
- **Employers** verify certificate authenticity

Technical Requirements

1. Network Setup

Set up a **simplified** Fabric network:

- 3 Organizations: University, Student, Employer
- 1 peer per organization
- 1 orderer (Solo is fine for this test)
- 1 channel: `education-channel`

Provided starter files: We'll provide basic network configuration files to save time.

2. Smart Contract Requirements

Required Functions (implement at least 5):

1. **IssueCertificate** - Only universities can issue
2. **GetCertificate** - Retrieve certificate by ID

3. **VerifyCertificate** - Verify authenticity (any org can verify)
4. **RevokeCertificate** - Only issuing university can revoke
5. **GetCertificatesByStudent** - Get all certificates for a student

Minimum Requirements:

- At least 6 test cases
- Cover success and failure scenarios
- Test access control
- Mock the transaction context

Integration Test Script (30 mins)

Create a simple Node.js script (`test-chaincode.js`) that:

1. Deploys the chaincode
2. Issues a certificate
3. Queries the certificate
4. Attempts unauthorized operations
5. Outputs results